# TQUK Functional Skills Qualification in Maths at Level 2

## Mark Scheme (Sample Assessment Paper 2)

**Mark scheme information**

This mark scheme is intended to support the valid and consistent marking of the examination paper identified above. This mark scheme includes:

- the total mark available for each question or sub-question
- the individual subject content coverage and mapping of each question or sub-question as well as coverage totals
- the marking process and considerations which could or should be followed
- the types of responses expected for each mark.

**Information for the marker**

- The pass **mark** for this paper is **36 marks.**
- This mark scheme documents covers both Section A (Non-Calculator) and Section B (Calculator).
- All marking must be completed consistently and the mark scheme must be applied fairly.
- Markers should award full marks if the candidate deserves full marks.
- Working is always expected, and space is provided for candidates to show their working.
- Questions where marks are awarded for working will always state 'show your working' or similar statement.
- Markers should be prepared to award zero marks if the candidate's response is not worthy of credit according to the requirements of the mark scheme for that question.
- For paper-based assessment, individual marks awarded to the candidate should be annotated clearly on the candidate's script. Once calculated and checked, overall marks achieved by the candidate must be included in the relevant area of the examination front cover.

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

## Glossary

| Marking Term | Definition |
|---|---|
| ACO | Accept only the correct answer |
| FOL | Follow-through marks are applied when there are earlier mistakes in the method |
| UNIT | The unit must be included in final answer for the mark(s) to be given |
| ALL | Identifies that all separate points must be met in order to receive full marks |
| NUM | Confirms that only the number is required, not the specific unit, type or measure |
| OE | Or equivalent |
| Coverage Term | Definition |
| UN | Use of number and the number system |
| UCM | Use of common measures, shape and space |
| HID | Handle information and data |
| PS | The ability to apply mathematical thinking effectively to solve problems |
| UPS | The ability to do maths when not as part of a problem |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

## Section A: Non-Calculator

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **1** | 1 | 1 | 1.022 | ACO | UPS | UN10i |
| **2** | 1 | 1 | 8.82 | ACO | UPS | UN10ii |
| **3** | 2 | 2 | 18 (mph) | **Award full marks if correct answer seen** | UPS | |
| | | 1 | 3 ÷ 10 (× 60) or 0.3<br>OR<br>3 × 6 | OE method | | UCM15i |
| | | 1 | 18 (mph) | ACO | | UCM15i |
| **4** | 2 | 1 | $\frac{86}{100}$ | ACO<br>OE fraction<br>ALL | UPS | UN4 |
| | | 1 | 86% | ACO<br>ALL<br>If zero scored, then award one-mark special case if their fraction and their percentage match each other | | UN4 |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **5** | 2 | 2 | (£)23.60 | **Award full marks if correct answer seen** | PS | |
| | | 1 | 20 × 0.18 or 3.6<br>OR<br>20 × 1.18 or 23.6 | OE method | | UN5i |
| | | 1 | (£)23.60 | ACO<br>Must be 2dp | | UN5i |
| **6** | 2 | 2 | 4625 | **Award full marks if correct answer seen** | UPS | |
| | | 1 | 2500, 3000, 4250, 5000 …<br>OR<br>6600, 5500, 5000, 4250 …<br>OR<br>(4250 + 5000) ÷ 2<br>OR<br>9250 ÷ 2 | OE method to work out median | | HID23i |
| | | 1 | 4625 | ACO | | HID23i |
| **7** | 2 | 2 | 125 000 cm OR 1.25 km<br>OR<br>1250 metres | **Award full marks if correct answer seen** | UPS | |
| | | 1 | 2.5 × 50 000 | OE method | | UCM18i |
| | | 1 | 125 000 cm OR 1.25 km | ACO<br>UNIT<br>OE correct distance with correct unit | | UCM18i |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **8** | 3 | 3 | $1\frac{1}{8}$ | **Award full marks if correct answer seen** | PS | |
| | | 1 | $6 - 3\frac{5}{8}$ or $2\frac{3}{8}$ or $\frac{19}{8}$<br><br>OR<br><br>$3\frac{5}{8} + \frac{10}{8}$ or $4\frac{7}{8}$ or $\frac{39}{8}$<br><br>OR<br><br>6 × 8 or 48<br>AND<br>3 × 8 + 5 or 29 | OE method or fraction<br>Accept use of decimals or percentages | | UN7ii |
| | | 1 | $2\frac{3}{8} - 1\frac{2}{8}$<br>OR<br>$\frac{19}{8} - \frac{10}{8}$ or $\frac{9}{8}$<br>OR<br>$6 - 4\frac{7}{8}$ OR<br><br>$\frac{48}{8} - \frac{39}{8}$<br><br>OR<br><br>48 – 29 – 10 or 9 | OE method or fraction<br>Accept use of decimals or percentages | | UN7ii |
| | | 1 | $1\frac{1}{8}$ | ACO | | UN7ii |
| **Total: 15 marks** | | | | | | |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

## Section B: Calculator

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **1** | 1 | 1 | –865<br>–820<br>–814<br>–811 | ACO | UPS | UN1 |
| **2** | 1 | 1 | 50 (inches) | ACO | UPS | UCM14i |
| **3** | 1 | 1 | 137.7 | ACO | UPS | UN12 |
| **4** | 1 | 1 | (–5, 3) | ACO | UPS | UCM19 |
| **5** | 2 | 2 | 40(°) | **Award full marks if correct answer seen** | UPS | |
| | | 1 | 360 – 120 – 120 – 80 | OE method | | UCM22i |
| | | 1 | 40(°) | ACO | | UCM22i |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **6** | 3 | 3 | Yes AND 4.3(332)(m) | **Award full marks if correct answer seen** | PS | |
| | | 1 | 3.14 × 1.38 | OE method | | UCM16ii |
| | | 1 | 4.3(332) | ACO<br>Implies 1st mark | | UCM16ii |
| | | 1 | Yes AND 4.3(332) (m) | Accept Yes AND Any correct reason.<br>FOL their 4.3(332) correctly compared with 4.5 if 4 < their 4.3(332) < 5 | | UN9 |
| **7** | 3 | 3 | No AND 71.5 ($m^2$)<br>OR<br>No AND 12 (litres) | **Award full marks if correct answer given** | PS | |
| | | 1 | 91 ÷ 14 or 6.5 ($m^2$ per litre) | OE method | | UN11ii |
| | | 1 | 91 ÷ 14 × 11<br>OR<br>78 ÷ (91 ÷ 14)<br>OR<br>78 ÷ 6.5 | OE method | | UN11ii |
| | | 1 | No AND 71.5 ($m^2$)<br>OR<br>No AND 12 (litres) | ACO<br>Accept No **AND** Any correct reason. | | UN11ii |
| **8** | 3 | 3 | Greenland AND –10 (°C) | **Award full marks if correct answer given** | PS | |
| | | 1 | $\frac{5(14-32)}{9}$ | OE method to substitute 24 into formula | | UN3ii |
| | | 1 | –10 (°C) | ACO | | UN3ii |
| | | 1 | Greenland AND –10 (°C) | Accept Greenland **AND** Any correct reason<br>FOL their –10 if their –10 < 0 | | UN1 |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **9** | 3 | 3 | (£)384 | **Award full marks if correct answer seen** | PS | |
| | | 1 | 33.6(0) ÷ 1.05 | OE method | | UN6b |
| | | 1 | 32 | ACO<br>Implies 1st mark | | UN6b |
| | | 1 | (£)384 | FOL the correct answer to their 32 × 12 | | UCM15iii |

| | | | | | | |
|---|---|---|---|---|---|---|
| **10** | 4 | 4 | No AND 89.6 | **Award full marks if correct answer seen** | PS | |
| | | 1 | 20 AND 60 AND 100 AND 140 | ACO Correct midpoints identified | | HID24 |
| | | 1 | (20 × 7) + (60 × 34) + (100 × 100) + (140 × 9)<br>OR<br>140 + 2040 + 10 000 + 1260 or 13 440 | Allow consistent use of upper of lower bounds multiplied by the frequency<br>Allow one error in midpoints, upper bounds or lower bounds | | HID24 |
| | | 1 | Their 13 440 ÷ 150 or 89.6 | FT their 13 440 from correct method<br>Allow consistent use of upper of lower bounds multiplied by the frequency divided by 150<br>Allow one error in midpoints, upper bounds or lower bounds<br>Do not allow 150 ÷ 4 | | HID24 |
| | | 1 | No AND 89.6 | Accept No **AND** Any correct reason | | HID24 |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

<table>
<tr><th>Q</th><th>Total Marks</th><th>Marks</th><th>Answer/Examples</th><th>Further Considerations/Comments</th><th>PS/UPS</th><th>SC</th></tr>
<tr><td rowspan="6">11</td><td rowspan="6">4</td><td>4</td><td colspan="2">Award full marks for fully correct table</td><td rowspan="6">PS</td><td></td></tr>
<tr><td>1</td><td>80 ÷ (5 + 3 + 2) or 8</td><td>OE method</td><td>UN11i</td></tr>
<tr><td>1</td><td>5 × their 8 or 40<br>OR<br>3 × their 8 or 24<br>OR<br>2 × their 8 or 16</td><td>OE method<br>FOL their 8 from correct method for ratio</td><td>UN11i</td></tr>
<tr><td>1</td><td>40 AND 24 AND 16 seen</td><td>ACO values do not need to be in correct position in table</td><td>UN11i</td></tr>
<tr><td>1</td><td>Fully correct table
<table>
<tr><th>Activity</th><th>Number of Students</th></tr>
<tr><td>Swimming</td><td>40</td></tr>
<tr><td>Rock climbing</td><td>24</td></tr>
<tr><td>Go karting</td><td>16</td></tr>
<tr><td>Total</td><td>80</td></tr>
</table></td><td>ACO</td><td>UN11i</td></tr>
</table>

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **12** | 4 | 4 | No AND (£)1 009 176(.75) | **Award full marks if correct answer seen** | PS | |
| | | 1 | 975 050 | ACO<br>May be seen or implied by subsequent working or correct answer | | UN1 |
| | | 1 | Their 975 050 × 0.035<br>OR<br>Their 975 050 × 1.035 | OE method<br>FOL their 975 050 if their 975 050 contains just the figures 9750 | | UN5i |
| | | 1 | 34 126.75<br>OR<br>1 009 176.75 | FOL the correct answer to their 975 050 × 0.035 or their 975 050 × 1.035 | | UN5i |
| | | 1 | No AND (£)1 009 176(.75) | FOL their 1 009 176.75 correctly compared with 1 010 000 | | UN11ii |
| **13** | 4 | 1 | 34 | ALL<br>ACO Mean | PS | HID25 |
| | | 1 | 18 | ALL<br>ACO Range | | HID25 |
| | | 1 | Yes AND correct reason<br>eg<br>mean score for week 2 is higher AND 34 seen | ALL<br>OE comment comparing means.<br>FOL their 34 if their 34 is in the range [22, 40] | | HID25 |
| | | 1 | No AND correct reason<br>eg<br>range for week 2 is higher AND 18 seen | ALL<br>OE comment comparing ranges.<br>FOL their 18 if their 18 is a result of subtracting two numbers from the table | | HID25 |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **14** | 5 | 5 | 13 (tins) | **Award full marks if correct answer seen** | PS | |
| | | 1 | 3.14 × 3 × 3 (× 2) or 28.26 or 56.52 | OE method to find the area of one or both circles | | UCM17ii |
| | | 1 | 3.14 × 3 × 3 × 2 + 2 × 3.14 × 3 × 12<br>OR<br>2 × 28.26 + 2 × 3.14 × 3 × 12<br>OR<br>56.52 + 2 × 3.14 × 3 × 12<br>OR<br>56.52 + 226.08 | OE method to work out total surface area | | UCM17ii |
| | | 1 | 282.6 ($cm^2$) | ACO<br>Implies first 2 marks | | UCM17ii |
| | | 1 | Their 282.6 × 50 ÷ 1150 or 12(.2869…) | OE method<br>FOL their 282.6 | | UN11ii |
| | | 1 | 13 (tins) | FOL their 12(.2869…) correctly rounded up to the nearest whole number | | UN9 |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| Q | Total Marks | Marks | Answer/Examples | Further Considerations/Comments | PS/UPS | SC |
|---|---|---|---|---|---|---|
| **15** | 6 | | **Alternative Method 1: Area of Triangle** | | PS | |
| | | 1 | 0.5 × 2.47 × 4.18 or 5.1623 | OE method to work out area of triangle<br>Accept any correct rounding or truncating of 5.1623 | | UCM16i |
| | | 1 | 0.5 × 2.47 × 4.18 + 5.65 × 4.18<br>OR<br>5.1623 + 5.65 × 4.18<br>OR<br>5.1623 + 23.617 | OE method to work out total area | | UCM16i |
| | | 1 | 28(.7793) (m) | ACO<br>Accept any correct rounding or truncating<br>Implies first 2 marks | | UCM16i |
| | | 1 | (£)5(.00) | ACO Mode | | HID23ii |
| | | 1 | Their 29 × their 5 | OE method<br>FOL their 29 and their 5<br>Their 29 must be their 28(.7793) correctly rounded up to the nearest whole number<br>Their 5 must be in the range [3.8, 6] | | UN9 |
| | | 1 | (£)145 | FOL the correct answer to their 29 × their 5<br>Their 29 must be their 28(.7793) correctly rounded up to the nearest whole number<br>Their 5 must be in the range [3.8, 6]<br>If pence given in the answer, it must be 2dp | | UCM15iii |
| | | | **Alternative Method 2: Area of Trapezium<br>Not expected at Level 2 but award if used** | | | |
| | | 1 | 0.5 × (8.12 + 5.65) or 6.885<br>OR<br>(8.12 + 5.65) × 4.18 or 57.5586 | OE method to start to use the formula for area of trapezium<br>Accept any correct rounding or truncating of 6.885 or 57.5586 | | UCM16i |

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 0.5 × (8.12 + 5.65) × 4.18 | OE method for the full formula for area of trapezium | | UCM16i |
| | | 1 | 28(.7793) | ACO<br>Accept any correct rounding or truncating<br>Implies first 2 marks | | UCM16i |
| | | 1 | (£)5(.00) | ACO Mode | | HID23ii |
| | | 1 | Their 29 × their 5 | OE method<br>FOL their 29 and their 5<br>Their 29 must be their 28(.7793) correctly rounded up to the nearest whole number<br>Their 5 must be in the range [3.8, 6] | | UN9 |
| | | 1 | (£)145 | FOL the correct answer to their 29 × their 5<br>Their 29 must be their 28(.7793) correctly rounded up to the nearest whole number<br>Their 5 must be in the range [3.8, 6].<br>If pence given in the answer, it must be 2dp | | UCM15iii |

**Total: 45 marks**

SAMPLE

 © Training Qualifications UK 2023 All Rights Reserved

## Mapping Matrix

| **Totals** | UN | UCM | HID | PS | UPS | SC |
|---|---|---|---|---|---|---|
| Section A | 9 | 4 | 2 | 5 | 10 | N/A |
| Section B | 21 | 15 | 9 | 39 | 6 | N/A |
| Total (%) | 50% | 32% | 18% | 73.3% | 26.7% | 20/28 |

## Ofqual Mapping Requirements

| | UN | UCM | HID | PS | UPS | SC |
|---|---|---|---|---|---|---|
| Total (%) | 45-55% | 30-45% | 10-20% | 73-77% | 23-27% | |

**End of Mark Scheme**

SAMPLE